You can see the result in Figure 6. Words like 'nature' are converted to 'natur', and so are words like natural.

Frequency distribution

The next process is frequency distribution, a method usually used to count the vocabulary. The output is a dictionary, which may have the vocabulary as the key and the number of times the word has occurred in the text as the value. This technique can be used in processes like sentiment analysis, where we can count the number of positive words, negative words, etc. This process is also used in vocabulary analysis and feature selection.

We can import the function using the following lines of code.

```
from nltk import FreqDist
fd = FreqDist(l)
print(fd.most_common(10))
```

```
'will',
'focu',
'on',
'the',
'basic',
'of',
'natur',
'languag',
'process',
'and',
'the',
'applic',
'of',
'these',
'topic',
'use',
'one',
'of',
'the',
'most',
'popular',
'nlp',
'librari',
'known',
'as',
'nltk',
```

Figure 6: The result

```
('has', 'VBZ'),
('become', 'VBN'),
('the', 'DT'),
('most', 'RBS'),
('popular', 'JJ'),
('topic', 'NN'),
('of', 'IN'),
('research', 'NN'),
('.', '.'),
('With', 'IN'),
('Generative', 'NNP'),
('AI', 'NNP'),
(',', ','),
('ChatGPT', 'NNP'),
(',', ','),
('we', 'PRP'),
('are', 'VBP'),
('moving', 'VBG'),
('more', 'JJR'),
('and', 'CC'),
('more', 'RBR'),
('towards', 'NNS'),
('NLP', 'NNP'),
('.', '.'),
('So', 'RB'),
('many', 'JJ'),
('new', 'JJ'),
('models', 'NNS'),
```

Figure 8: Parts of speech

NER and has multiple business applications. We can use it to detect private data in text, as well as extract important names/ dates, a news article or a normal piece of text. We can use this process for question answering, document summarisation, and more. The main objective of NER is to identify names, locations, dates, and more. Now let us see how we can use this. First, import NER using the following lines of code:

```
text = "Jishnu is an amazing person"
tokens = nltk.word_tokenize(text)
tagged = nltk.pos_tag(tokens)
entities = nltk.chunk.ne_chunk(tagged)
print(entities)
```

As we can see in Figure 7, the counts of each of the words are given in the form of a dictionary.

As you can see in in the output in Figure 9, the proper nouns, nouns, etc, have been identified correctly.

```
[15] from nltk import FreqDist

[23] fd = FreqDist(l)
     print(fd.most_common(10))

     [('the', 5), ('.', 5), (',', 5), ('of', 4), ('we', 3), ('more', 3), ('and', 3), ('In', 2), ('world', 2), ('Natural', 2)]
```

Figure 7: The count

Tagging parts of speech, and chunking

Another process that is important is tagging the parts of speech, which must be done accurately. Here we tag the parts of speech for each word in the given piece of text. Let us see how we can do that!

We can get the parts of speech using the following line of code. The same list of tokens is being used here.

```
nltk.pos_tag(l)
```

Another important process in the field of NLP is chunking. This is done to identify phrases, or groups of words that are generally used together, such as 'a piece of cake'. We can use regex to identify these phrases.

Named entity recognition

The next process we are going to talk about is a very interesting one. This is called 'named entity recognition' or

```
text = "Jishnu is an amazing person"
tokens = nltk.word_tokenize(text)
tagged = nltk.pos_tag(tokens)
entities = nltk.chunk.ne_chunk(tagged)
print(entities)

(S (GPE Jishnu/NNP) is/VBZ an/DT amazing/JJ person/NN)
```

Figure 9: Final output

That's it! You have learnt most of the basic concepts in the field of NLP. You can explore much more in this space as there are so many libraries today helping with various NLP tasks. END

By: Jishnu Saurav Mittapalli

The author is a full-stack web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.